

Semantic Structure and Revision in Temporal Evidence Graphs for Generative Monitoring

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Abstract. Generated explanations can remain plausible after their evidence changes. We study a typed temporal property graph that separates recording provenance, model-proposed propositions, declared prerequisites and time-indexed support. A query-specific semantic projection preserves numerical witnesses and version history while exposing the dependencies relevant to revision. A shared, labeled network view makes immutable citations, changed versions and alternative witnesses inspectable in the paper and review dashboard. Two conditional results connect local support re-evaluation to transitive exposure: the fraction of necessary changes excluded by direct-only scheduling. Re-analysis of synthetic, WESAD and PPG-DaLiA experiments finds that only 18 of 1201 displayed claims have parents; every eligible waveform revision has zero transitive exposure. Independent semantic scoring separates source grounding from target completeness: checked displays are fully grounded, but subject-averaged required recall is 75.5–78.2%. In 240 controlled structural cases, exact and GPU-generated programs expose necessary intermediate dependencies and surviving alternative witnesses. Exposure predicts direct-maintenance misses in all 860 eligible replays; full maintenance agrees across Neo4j and an indexed relational control. Generated answers realize evidence-to-answer depth one or two despite requests up to four. These results identify when explicit semantic structure enables selective revision, and show why storage size, path length and contract acceptance alone cannot establish explanation fidelity or maintenance benefit.

Keywords: Temporal knowledge graphs · Evidence provenance · Generative explanations · Incremental maintenance · Physiological streams

1 Introduction

A generated explanation can become misleading when a sensor window is corrected, a feature fails a quality check, or a delayed observation replaces a statement of absence. Deleting the answer discards useful information; retaining it can preserve a wrong number. A temporal evidence graph exposes what was asserted, which observations and prerequisites were cited, and what was known when.

A comparison may declare a parent while also citing both operands directly; revising either operand then reaches it in one step. Conversely, a conjunction supported only through an intermediate claim can remain displayed after a direct checker updates that intermediate. A third case reaches a statement but preserves

it through another admissible witness. Edge semantics and support predicates, rather than visual complexity, distinguish these regimes.

We study a researcher-specified, typed temporal property graph instantiated from recording provenance and generated propositions. The contribution has three parts. First, we define its construction and a query-specific semantic projection with mappings back to stored objects and a shared labeled view for a human reviewer. Immutable citations, version changes and grouped support witnesses are visible in both the paper and an evidence-review dashboard. Second, we connect an explicit support model to a locality result and a measurable condition for direct/full maintenance equivalence. Third, we evaluate that connection using actual Neo4j exports, independent proposition scoring, and a controlled structural extension of three completed benchmarks. The graph exposes commitments made by the generated answer; it is not a measurement of internal neural reasoning.

The completed waveform study supplies a useful contrast. Only 18 of 1201 M1 displayed claims declare parents, and direct checking matches transitive maintenance. Our structural analysis finds zero transitive exposure in every eligible waveform revision batch. A new protocol also distinguishes accurate background statements from violations of a strict target-only output contract. This exposes the tradeoff between grounded content and required-fact coverage. Finally, exact numerical programs and common GPU-generated candidates vary dependency depth and admissible alternative witnesses while preserving the underlying recordings. The resulting evidence identifies when propagation matters and when simpler maintenance suffices.

1.1 Relation to Prior Work

Truth-maintenance systems revise justified beliefs when premises change [1]; incremental view maintenance updates database results from changes to their inputs [4]. Provenance semirings distinguish jointly required inputs from alternative derivations [3]. Our contribution connects these established ideas to dependencies actually expressed by a bounded generator and independently measured maintenance outcomes.

PROV-DM distinguishes entities, activities, derivation, revision and attribution [13]. It informs our conceptual separation of recording provenance, generated assertions and review events. The implementation remains a property graph with Python admission checks and database identity constraints. GraphRAG constructs entity graphs and community summaries for corpus-level question answering [2]; Graphiti, described in the Zep architecture, maintains temporal information for agent memory [6]. Here the unit of evaluation is a revisable numerical proposition and its admissible witnesses. This distinction also separates source-grounded semantic fidelity from the faithfulness of a model’s account of its own computation, which need not match its actual decision process [11].

2 A Typed Temporal Semantic Representation

2.1 Schema-Guided Ontology Instantiation

The ontology is fixed in code; the generator proposes instances inside it. Formally, let

$$\begin{aligned}\Omega &= (\mathcal{T}, \mathcal{R}, \text{dom}, \text{ran}, \mathcal{C}), \\ G_t &= (V_t, E_t, \vartheta_V, \vartheta_E, a_t).\end{aligned}\tag{1}$$

Here $\mathcal{T}$ and $\mathcal{R}$ are node and relation vocabularies; $\vartheta_V : V_t \rightarrow \mathcal{T}$ and $\vartheta_E : E_t \rightarrow \mathcal{R}$ give their types. An edge of type r connects allowed types in $\text{dom}(r)$ and $\text{ran}(r)$. Attributes a_t include immutable and logical IDs, dataset/subject/session scope, event intervals, ingestion time, version, value, unit and evidence state. Constraints $\mathcal{C}$ include immutable identifiers, finite acyclic declared dependencies and the admission predicates below. Ingestion time is written τ_v , distinct from the type maps ϑ . At knowledge time t , only records with $\tau_v \leq t$ are visible. Current assessments use their recorded knowledge time for visibility. Current queries resolve each logical identity to its greatest visible version, breaking ties by ingestion time and ID; historical queries retain their specified knowledge time. Tied assessments retain their trigger IDs; conflicting states are marked ambiguous because the stored timestamp alone does not order triggers.

Table 1 identifies stored objects. Windows carry hashes, sample indices and channel metadata; raw arrays remain outside the database. Features carry units, extractor metadata and source IDs. The `with_entities` function adds subject and channel identities. The model proposes proposition fields, evidence references, parent links and sentences. Checks admit bounded claims before immutable insertion. Neo4j additionally creates `SupportAssessment` nodes and `ASSESSED_AS` edges, included in storage accounting. Source-endpoint insertion can create an unresolved Record placeholder; none occurs in the original exports. Table 2 gives relation meanings and construction rules. Neo4j enforces unique scope/ID pairs; Python checks record kinds, immutable payloads, acyclicity and numerical admission. The database does not independently enforce those semantic predicates.

The construction can be written as

$$G_t = \Gamma_\Omega(E_{\leq t}, \hat{C}_{\leq t}) = \text{Insert}_{\text{immutable}}\left(\text{Check}_q\left(\text{Normalize}_\Omega(E_{\leq t}), \hat{C}_{\leq t}\right)\right).\tag{2}$$

$E_{\leq t}$ is the observation/revision log; $\hat{C}_{\leq t}$ contains model proposals generated from the normalized visible evidence. Rejected proposals remain in the request archive. Inserted assessments and explanation versions extend the instance without overwriting earlier content. This is schema-guided instantiation, rather than ontology discovery.

Construction determines much of the apparent topology. Each `source_ids` entry creates dependent-to-input `DEPENDS_ON`; features also receive `DERIVED_FROM`, and accepted claims the opposite-direction `SUPPORTS` mirror. Display membership

Table 1. Implemented object vocabulary. Checked status describes an admission or assessment operation; it does not establish physiological truth.

Stored object	Meaning	Construction
Subject / Sensor	Identity and channel	deterministic
Observation	Hashed sample window	deterministic
FeatureVersion	Value or availability watch	deterministic
ClaimVersion	Bounded proposition, references	model; checked
ExplanationVersion	Text and display membership	administrative
ReviewEvent	Recorded review action	administrative
SupportAssessment	State, time and trigger	evaluation
Record placeholder	Unresolved referenced ID	administrative

Table 2. Relation construction and direction. R denotes a Record, P an unresolved placeholder, F a feature, C a claim, O an observation, X an explanation and A an assessment. The adapter constructs edges; admission checks supply claim-specific semantics. Administrative relations do not express entailment.

Relation	Domain $\rightarrow$ range	Meaning and creation rule
DEPENDS_ON	$R \rightarrow R/P$	Each source ID: deterministic input, model citation or declared parent.
DERIVED_FROM	$F \rightarrow R/P$	Feature source IDs: extraction provenance.
SUPPORTS	$R/P \rightarrow C$	Reverse source edge on admission; validity at insertion only.
SUPERSEDES	$R \rightarrow R/P$	New version to its predecessor ID.
CONTAINS	$X \rightarrow R/P$	Explanation source IDs: display membership; also dependencies.
ASSESSED_AS	$R \rightarrow A$	Outcome keyed by scope, record, knowledge time and trigger.
CONTRADICTS	$R \rightarrow R$	Trigger to adversely assessed record; time from assessment.
FOR_SUBJECT	$R \rightarrow \text{subject}$	Subject membership from deterministic identities.
OBSERVED_BY	$O \rightarrow \text{sensor}$	Channel identity from recording metadata.
REVIEWED_BY_EVENT	$R/P \rightarrow \text{review}$	Reviewed item to appended review event.

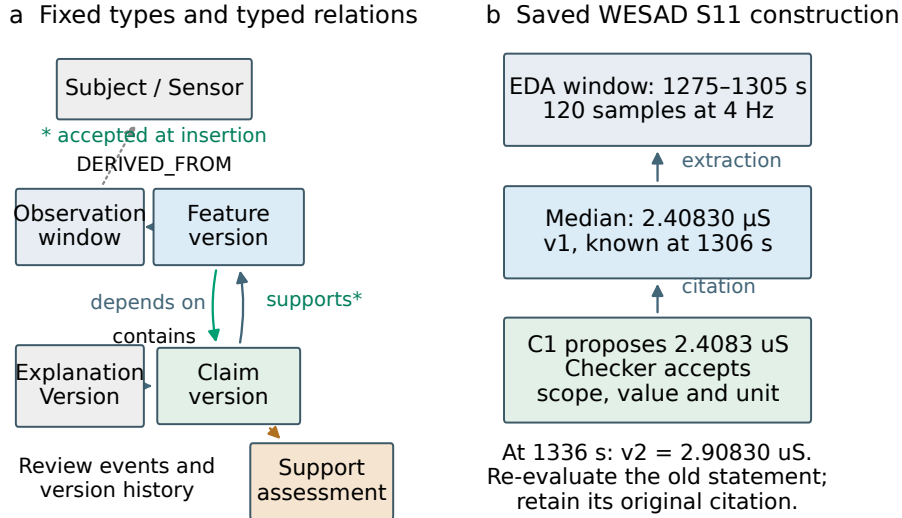


Fig. 1. Fixed vocabulary and actual instantiation. Blue denotes deterministic recording/feature construction, green a checked model proposition, grey identity or display administration, and amber assessment/version effects. The concrete path comes from the exported WESAD S11 correction scope; its source hash, sample indices and full IDs accompany the figure. Arrows have the stored direction. The displayed schema is a readable subset of the complete construction table.

adds two relations: **CONTAINS** and **DEPENDS_ON**. Subject links create **FOR_SUBJECT** hubs. We separate these administrative, provenance and reasoning roles.

In the saved WESAD S11 episode (Figure 1), the EDA window at 1275–1305 s produces a median of 2.408303 μ S. The model proposes 2.408303 μ S and cites that feature. The checker admits the bounded statement and insertion records both its **DEPENDS_ON** citation and the reverse **SUPPORTS** edge. A later feature version changes the median to 2.908303 μ S, retaining the original observation hash and immutable claim citation. Re-evaluation changes the claim’s current support without rewriting its history.

2.2 Edges as Explicit Semantic Commitments

A citation identifies a proposed evidence witness. A parent link declares a prerequisite. A deterministic extraction edge records how a feature was produced. Membership and co-occurrence carry neither implication. In particular, an LLM-proposed **DEPENDS_ON** edge does not prove that its target is a necessary premise. **SUPPORTS** records acceptance at insertion, not support at every later time. A **CONTRADICTS** edge records an adverse assessment triggered by a revision; its associated assessment supplies the knowledge time. Exports retain these origins and the distinction between proposed, accepted and displayed links.

2.3 The Query-Specific Semantic Core

We define a projection rather than discover a second ontology:

$$K_{q,t} = \pi_{q,\Omega}(G_t), \quad V(K_{q,t}) = C_{q,t}^* \cup F_{q,t}, \quad E(K_{q,t}) = D_{q,t} \cup P_{q,t}. \quad (3)$$

$C_{q,t}^*$ contains the latest displayed answer’s member claims and their declared claim ancestors. $F_{q,t}$ contains their visible cited feature propositions, including explicit availability watches. $D_{q,t}$ retains the corresponding stored dependency edges; $P_{q,t}$ contains labelled derived paths that resolve an immutable citation through a visible version chain. Each derived path retains its source citation and supersession witness IDs. It is never drawn as a directly stored Neo4j relationship.

Subject/sensor identity, explanation containers, assessments and old versions are sidecars. Feature-to-observation provenance is retained there and may be expanded in a drawing. Claim scope comes from the asserted proposition fields, rather than the storage wrapper’s query interval. Full IDs, raw sentences, original versions and assessment times remain recoverable. Thus the projection preserves the explicit commitments needed for the answer and its revision; its smaller size is a consequence of a specified selection rule.

We distinguish the complete stored graph, an active evidence-and-claim projection at t , and $K_{q,t}$. The active projection includes latest evidence and current display members across queries, retaining actual observation provenance. The core answers more focused competency questions: which window supports a value, which operands support a comparison, which version was known, what changed after correction, and whether an alternative witness survives. We evaluate answers to these questions against immutable input logs or exact program specifications, independently of database support flags.

2.4 Producing the Human-Review Network View

The review view expands the semantic core for a selected answer or focus claim. It restores original cited versions, required observation provenance and visible successor versions instead of redirecting immutable citations. For a stored-node budget B and selected recorded times T , the implemented transformation is

$$H_{q,t} = \text{View}_B(G_t, q), \quad p = \text{Layout}\left(\bigcup_{u \in T} H_{q,u}; m\right), \quad I_{q,t} = \text{Render}(H_{q,t}, p). \quad (4)$$

H is a bounded review expansion of K , not another learned representation. p stores node coordinates and routed edges. Mode m selects deterministic Graphviz `dot` for structured views or seeded spring positioning for the classic network. The latter uses an undirected simple projection only to position nodes; analysis and arrows retain the typed directed graph. Collision separation and routing adjust presentation, not support semantics. A comparison uses the union layout and reuses p ; records not yet ingested remain absent from the earlier drawing. Coordinates and Euclidean distances have no semantic-similarity or neural-state interpretation.

The implementation selects database scope, answer, focus and knowledge time, follows declared inputs, expands feature-to-observation paths on request, and filters versions and assessments by their ingestion/knowledge times. Event intervals describe what was measured or asserted, separately from when it was known. Original `DEPENDS_ON` edges remain intact; current-version resolution retains its citation and reverse-supersession witnesses in an explicit derived-path sidecar. Availability, support assessment and display membership are separate fields. Of 432 tied record/time assessment groups, 245 contain conflicting states: all outcomes and trigger IDs remain visible as ambiguous, while membership follows the saved explanation version. These flags and drawings never supply evaluator reference truth.

The budget counts stored records, preferring complete alternative-witness closures when they fit. Missing records and incomplete witness groups trigger a *partial view* label and expansion control; the drawing does not claim that omitted context is irrelevant. Subject/sensor hubs, display containers, assessment nodes and reverse `SUPPORTS` mirrors stay in details. For multi-input witnesses in structured mode, labeled AND enclosures group feature rows: each row maps to its stored ID, and a dashed dependency bundle enumerates its original edges. Enclosures and bundles are rendering constructs. Between witness sets support is OR; within each set all inputs are required.

Green/blue/grey fills distinguish claims/features/observations; outlines and explicit words distinguish revisions, support and withdrawal. Every arrow is labeled and preserves stored direction or identifies a grouped rendering edge. The same typed JSON view, ID maps and Graphviz renderer produce Figures 2 and 3 and the existing Streamlit dashboard. Its optional *Classic network* mode draws Figure 4 from the same typed objects as individual ellipses. Selectboxes choose a recorded case, time, focus, witness route and node or edge; details recover full sentences, exact values/units, intervals, versions, source IDs and assessment provenance. Highlighting changes appearance only. Expansion exposes omitted provenance; review actions append events in an isolated workspace. This implements inspectable external commitments, without claiming a measured human-performance benefit or access to neural reasoning.

3 Support, Locality and Transitive Exposure

3.1 Support Through Admissible Witnesses

For a proposition c , let $\mathcal{W}(c)$ be a finite family of admissible witnesses. Each W is a set of jointly required evidence or prerequisite propositions; different W are alternatives. We use

$$\text{Supported}(c, t; q) = \bigvee_{W \in \mathcal{W}(c)} \left[\psi_c(W, t; q) \wedge \bigwedge_{u \in W} \text{Valid}(u, t; q) \right]. \quad (5)$$

Valid requires observable availability and admissible temporal and subject scope; for a prerequisite claim it recursively requires support. ψ_c checks the asserted

interval, selected logical versions, quantity, normalized unit and numerical operation. Numerical agreement uses the frozen tolerance $|v - \hat{v}| \leq \max(10^{-6}, 0.01|\hat{v}|)$. A correction can therefore invalidate ψ_c while leaving the source available. A supported historical assertion at its original knowledge time and a supported current assertion are different propositions. The extension’s specified inequalities use exact floating-point comparisons.

The original model supports local AND/OR composition and a two-operand difference. Generated numerical claims require all declared parents plus an admissible direct numerical witness; comparisons cite both operands. The extension uses explicit OR lists of AND inputs composed through an acyclic positive claim graph, not unrestricted Boolean formulas. Missing-evidence statements require a bounded producer universe and explicit availability watches; absence in an open graph is insufficient. The extension stores witness grouping in each record’s semantic-program metadata; source IDs contain their union for traversal. Edge presence alone does not recover the AND/OR grouping.

As in provenance and truth maintenance [3,1], losing one route preserves support when another admissible witness survives. Our controlled alternatives use disjoint recording windows, not renamed copies of one measurement. Source independence does not imply statistical independence.

3.2 Locality of a Revision

Let U_δ contain revised input identities, all earlier versions that can still be cited, and any changed availability subscriptions. In the dependency projection, edges point from dependent to input. Define

$$R_\delta = U_\delta \cup \text{RevReach}(U_\delta), \quad B_\delta = \{c : S_{t-}(c) \neq S_{t+}(c)\}, \quad (6)$$

where $S_t(c)$ abbreviates Equation 5. R_δ is a candidate set; B_δ is the actual support-change set.

Proposition 1 (Conditional locality). *For a finite acyclic dependency model, $B_\delta \subseteq R_\delta$ when every mutable input to each support predicate is explicitly represented and only U_δ changes between the two snapshots.*

Proof sketch. Order nodes from inputs to dependents. A node outside R_δ has no changed input and no prerequisite in R_δ ; otherwise it would be reverse reachable. Its scope, numerical operands and availability state are therefore unchanged. Induction in the acyclic order gives equal prerequisite support and equal predicate values, hence equal support. Considering all earlier versions covers immutable citations to v_1 after v_2 or v_3 arrives. Explicit availability records cover an arrival that falsifies an absence statement. $\square$

Locality fails for unrecorded clock expiry, newly discovered unsubscribed witnesses or external predicate changes. Our fixed-target queries and controlled witness universes represent changes through version/availability events. Benign updates and surviving alternatives demonstrate why reachability is only a candidate set, with potentially strict inclusion $B_\delta \subset R_\delta$.

Indexed adjacency traversal costs $O(|V_{R_\delta}| + |E_{R_\delta}|)$, plus current version lookups and witness evaluation. Actual costs also include snapshots, serialization, database round trips and writes. The original runtime repeatedly constructs version maps; structural replay evaluates the whole small program before scheduled writes. Neo4j variable-length queries may enumerate paths; SQLite uses a recursive CTE with duplicate elimination. Neither implementation inherits a linear-time guarantee from the abstract traversal bound.

3.3 When Direct Maintenance Suffices

Let A_δ be the independently determined set of prior admitted claims requiring a state change, and D_δ the claims actually scheduled by direct-only maintenance. For $|A_\delta| > 0$, define

$$\xi_\delta = \frac{|A_\delta \setminus D_\delta|}{|A_\delta|}. \quad (7)$$

We report NA when A_δ is empty. The primary measured exposure concerns withdrawal of initially grounded claims; restoration is recorded separately as a support transition. A_δ comes from the event-log/task oracle, not from an accepted flag or the existence of a graph path.

Proposition 2 (Direct/full equivalence and misses). *Assume initially correct states, complete re-evaluation of D_δ , dependency semantics consistent with the oracle, and no unrelated refresh or other mechanism that revisits excluded claims. Before fresh generation, direct maintenance misses the fraction ξ_δ of required changes. A sufficient condition for direct and full maintenance to agree is that every changed support predicate is directly scheduled. In particular, this holds when each changed input affecting an answer has a direct dependency edge to every affected claim and all other predicates remain unchanged.*

Proof sketch. Complete direct re-evaluation changes every required member of D_δ . Excluded members retain their initial state by the no-refresh assumption, so the missed set is exactly $A_\delta \setminus D_\delta$. If all changed predicates are scheduled, both algorithms compute the same predicates from the same snapshot; the remaining predicates retain equal states. $\square$

A nontrivial counterexample uses c_1 = “test 1 holds” and c_2 = “tests 1 and 2 hold”, with c_2 supported by c_1 and test 2. Correcting test 1 to false requires changing both claims. A direct scheduler reaches only c_1 , giving $\xi_\delta = 1/2$. If c_2 also directly cites test 1, it is directly scheduled and exposure becomes zero. If the answer has another independently sufficient route, losing the first route need not change the answer at all. Thus path length alone predicts neither a miss nor a withdrawal. The waveform exports and the controlled structural experiment test these distinct regimes.

4 Evaluation Design

4.1 Preserved Three-Source Study

The unchanged `minimum_v1` study uses synthetic physiological-style waveforms, WESAD laboratory recordings [10,9], and PPG-DaLiA daily-activity recordings [8,7]. Each contributes ten held-out subjects with two episodes each. Real-source splits use PCG64 seed 20260928 and five separate development subjects. Physiological/activity labels enter neither generation nor checking; outcomes concern recorded features.

Each episode has a 30-second target and a preceding 120-second comparison interval. Four replay variants provide clean/benign, channel-fault, delayed, and numerical-correction events; three knowledge checkpoints occur at target end plus 1, 31 and 91 seconds. Four question families require the target EDA median plus either its preceding-window difference, EDA missing fraction, or acceleration variability. These requirements are fixed independently of answers. Five methods produce 3,600 initial cases. B0 displays unchecked proposals; B1 performs direct maintenance over indexed records; B2 uses Neo4j with direct maintenance; M1 uses Neo4j with transitive maintenance; B3 uses indexed SQLite and recursive CTEs with the same transitive support semantics. All checked methods share fresh-answer checks and at most one repair attempt.

Pinned Qwen3-8B BF16 [5] originally uses a three-claim contract and 1,536 output tokens. Both studies use the existing RTX 5090, verified CUDA parameters and forward tensors, and vLLM structured decoding [12]. Database/statistical work uses the host CPU; Neo4j Community 5.26 and indexed SQLite retain their adapters. The three original B0 transport failures remain empty outputs.

4.2 Independent Semantic Evaluation

The new `semantic_analysis_v1` oracle resolves values and witnesses from immutable events and task specifications, never support flags or a neural verifier. Canonical identity includes dataset, subject/session, asserted interval, quantity, operation, normalized value/unit, temporal mode and logical version. Correctly labelled background can be grounded without meeting a target requirement. Wrong numbers, subjects and target attributions still fail. Citation/witness fidelity is separate from directly checkable proposition truth.

For emitted checkable propositions $C(q, t)$ and answerable required propositions $R(q, t)$, we report

$$\begin{aligned} P_{\text{grounded}} &= \frac{\sum_{c \in C} \mathbf{1}[\text{source-grounded } c]}{|C|}, \\ R_{\text{required}} &= \frac{|\{r \in R : \text{matched by a grounded } c\}|}{|R|}. \end{aligned} \tag{8}$$

Every requirement is matched once, so repeated paraphrases cannot increase recall. The optional harmonic summary $S = 2P_{\text{grounded}}R_{\text{required}}/(P_{\text{grounded}} +$

R_{required}) combines grounding and task completeness; their universes differ, so it is not conventional set F1. Empty output has precision NA and zero recall/summary when the question is answerable. Fully unanswerable questions have recall/summary NA and separately checked absence statements. Partially answerable questions contribute their answerable facts.

We separately score raw and final proposals, accepted claims and displays. Raw text, literal discrepancies and additional interpretation terms remain in a limited prose audit; structured scores do not certify unrestricted language. Temporal integrity reuses each prior answer before arrival, after arrival and after maintenance. These are discrete checkpoints, not recovery curves. Withdrawal and adequate replacement are separate outcomes.

4.3 Controlled Structural Extension

`structure_study_v1` selects the first five naturally sorted existing held-out subjects per source and both episodes, before observing new outputs. The 30 episodes reuse participants; they are not a new independent cohort. Four intended evidence-to-answer depths (1–4) and two support regimes yield 240 cases. Each route applies four explicit numerical comparisons to EDA median, EDA IQR, acceleration SD and EDA missing fraction. Bounds are fixed from the clean episode so the routes initially satisfy the tests. The answer states whether at least one eligible window meets all four criteria. A second route uses the disjoint preceding window, providing a genuinely different observation witness for this existential statement. The task makes no physiological diagnosis.

Depth changes grouping while preserving the tests: direct inputs; a complete route then answer; a two-test prefix then route and answer; or first-test, two-test prefix, route and answer. Exact compiled fixtures instantiate these programs. Separately, the model proposes up to 16 claims and explicit witness lists, with a 4,096-token output allowance and at most one repair. Model links are never filled in by the compiler. Requested and realized depths, omitted roots and incomplete alternative-witness families are retained. Two development pilots precede the core freeze. The second clarifies readable sentences and reuse of emitted parent IDs through a generic grouping example; neither pilot inserts required links. Both use development episodes. Their 40 calls are separate from the core budget.

One common admitted candidate per case is replayed through all maintenance methods, with no post-correction generation. B0 here is a maintenance-off counterfactual on that same admitted content. Four independent update conditions make the first test false in every route, remove only the primary route, make a benign numerical change, or revise an unused cardiac feature. Neo4j and relational methods receive identical candidates, events and predicates. An independent truth-table evaluator enumerates the bounded Boolean assignments to test witness entailment and equivalence, allowing different valid derivations. A separate event-log evaluator determines the actual required changes.

4.4 Graph Measurement and Uncertainty

Read-only Neo4j exports preserve nodes, relations, assessment times, Cypher and element IDs. Counts match the original ledger and sampled stored paths. We measure full storage, active graphs and query cores separately: typed mixing, degree/depth, direct citations, witness counts, reach and actual changes. Dependency degree excludes SUPPORTS and provenance mirrors; meaningful zero categories remain. No power-law or embedding interpretation is imposed.

Snapshots and variants are first aggregated within episodes, then episodes within subjects. Pointwise 95% intervals use 2,000 subject-cluster bootstrap resamples with seed 20260929. Historical versions and replay copies are never independent participants. Detailed distributions and exhaustive operational logs remain repository data. All definitions and primary denominators needed to interpret the results are retained here.

5 Results

5.1 Storage Structure and Semantic Structure Differ

The read-only export covers 480 retained Neo4j scopes; counts match the original ledger and 1850 representative stored paths are verified. M1 alone stores 47,038 nodes and 128,789 relations across its replay scopes. These are storage totals, not independent observations. Membership and extraction rules contribute much of the edge mass through their construction rules. Neither those edges nor the accepted-citation mirror demonstrate additional reasoning depth.

The query core averages 2.8–4.1 nodes across sources, or 1.5–2.2% of visible storage. This reduction follows the fixed selection of answer propositions and declared inputs; the active graph also includes other visible evidence. Observation witnesses remain recoverable through sidecars. The full stored totals count historical versions, whereas core sizes count visible propositions per query; subject-level aggregation prevents replay copies from becoming independent samples.

Only 18 of 1201 displayed M1 claims declare a parent: 98.5% are parentless. Claim-parent depth never exceeds one in the displayed waveform cores. Figure 2 illustrates why even that parent edge need not create transitive exposure: the comparison cites the revised target feature directly. Across 360 revision batches, 210 require a withdrawal and 0 have nonzero exposure; the locality audit finds 0 violations. This structure explains the completed direct/full equivalence without attributing it to an absence of stored paths or to a particular database engine.

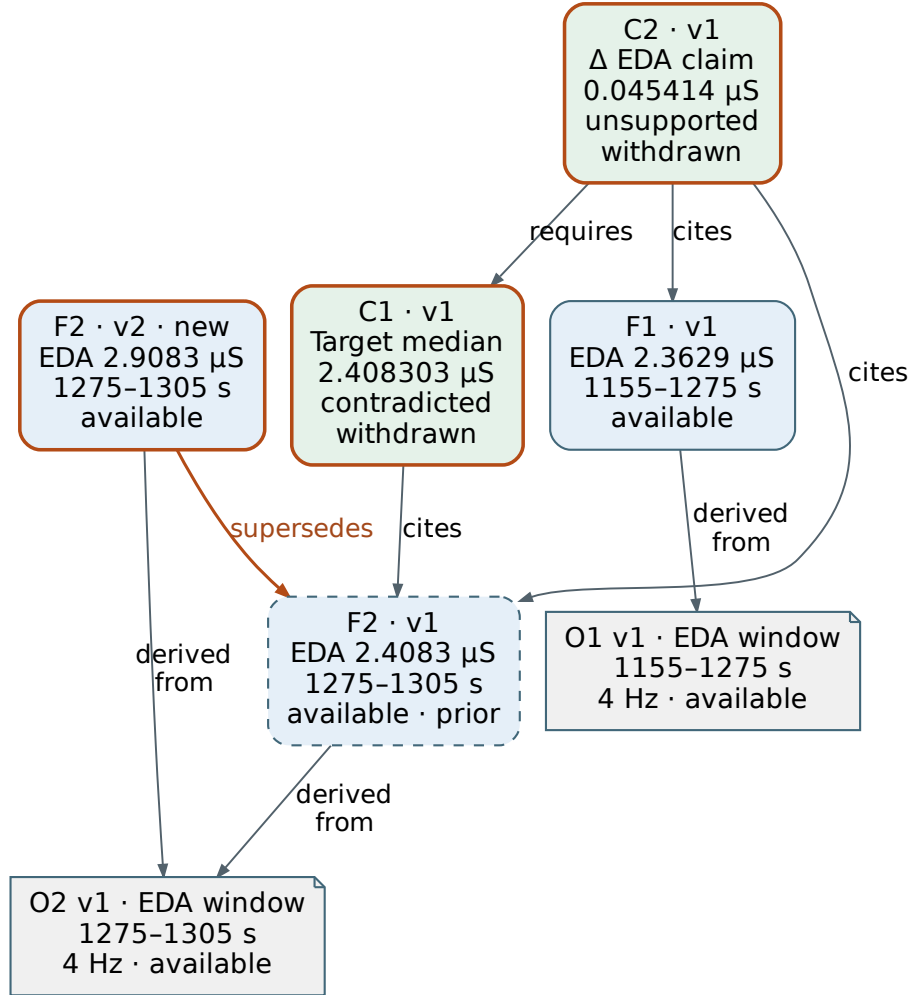


Fig. 2. Human-review network from actual Neo4j records: WESAD S11, episode 0, knowledge time 1336.01 s. The unchanged median-size rule selects this case among 20 eligible correction cores. Green nodes are claims, blue features, folded corners observations. *Cites/requires* mean `DEPENDS_ON`; *derived from* means `DERIVED_FROM`; *supersedes* is the stored new-to-old relation. The injected correction adds F2 v2; C1/C2 keep their original v1 citations and are withdrawn. Measurement availability remains distinct from current support. C2 directly cites both operands, so direct checking suffices here. Identity hubs, admission mirrors and assessment/display containers are hidden; their details remain inspectable. Labels round values; metadata preserves full precision.

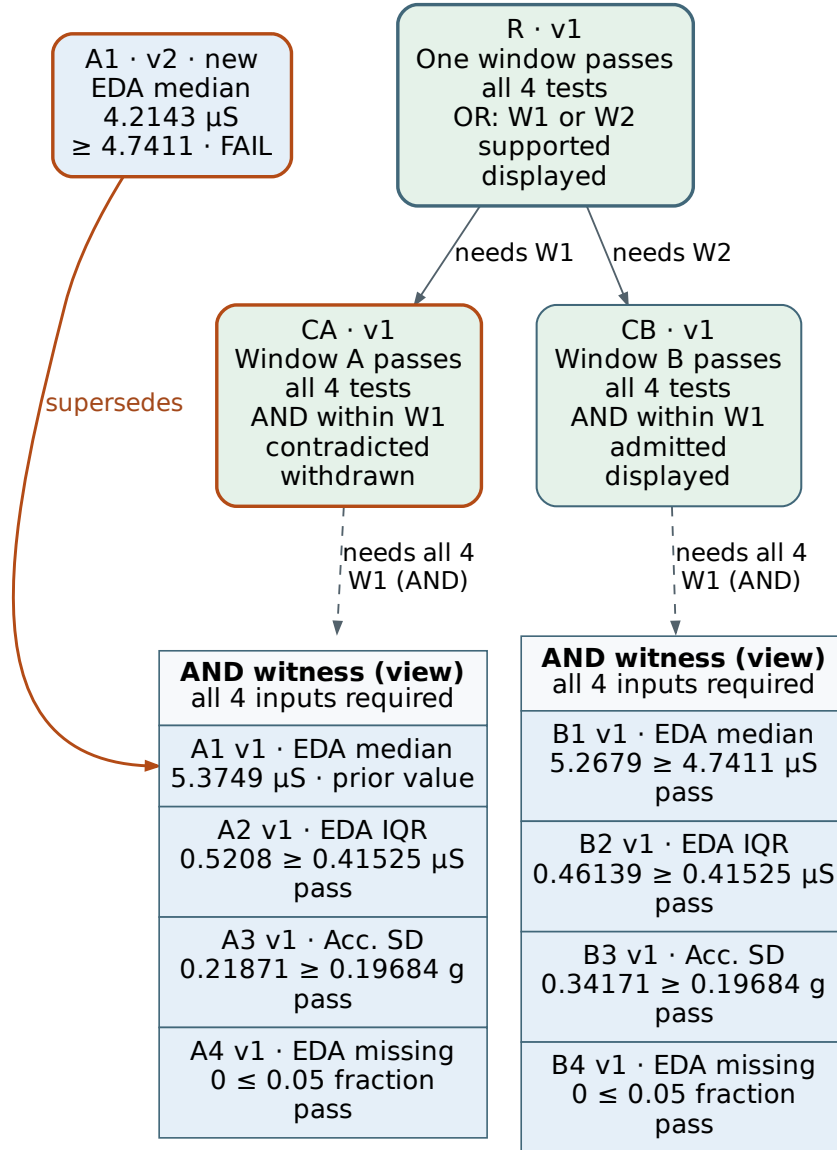


Fig. 3. Actual Neo4j generated-program view, PPG-DaLiA S1 episode 0, knowledge time 2331.01 s; requested and realized depth 2. This is the smallest complete eligible graph, with ID tie-breaking, among 87 programs. Windows A/B span 2270–2300/2150–2270 s. A1’s correction invalidates CA; CB still supports R. W1/W2 are alternatives at R; each route requires all four inputs. Blue rows are stored feature records. Dashed bundles summarize four `DEPENDS_ON` edges; AND enclosures are rendering constructs with exported row/edge mappings. Solid *needs* arrows are stored parent dependencies. Observation provenance is expandable; administrative details remain inspectable.

A worked ontology instance. Figure 4 exposes the same admitted PPG-DaLiA program as individual instances of the vocabulary in Equations 1–2: A1 is a particular **FeatureVersion**, not the abstract feature class. The drawing contains 14 records and 18 stored relations; the full scope contains 26 objects and 83 relations. Its 11 starred records form the current core (Equation 3). Withdrawn CA remains a declared ancestor of displayed R; it is not itself displayed. The old A1 and two EDA windows are added review context. Motion windows remain expandable provenance.

For this saved program, Equation 5 instantiates as

$$\begin{aligned} S(C_A, t) &= b_{A1}(t) \wedge b_{A2}(t) \wedge b_{A3}(t) \wedge b_{A4}(t) \\ S(C_B, t) &= b_{B1}(t) \wedge b_{B2}(t) \wedge b_{B3}(t) \wedge b_{B4}(t) \\ S(R, t) &= S(C_A, t) \vee S(C_B, t) \end{aligned}$$

Each b_{ri} requires the latest visible feature for PPG-DaLiA S1, session 1, the specified A/B interval, quantity and unit, with an available non-null value. It then applies the task’s exact $\geq$ threshold for median, IQR and acceleration SD, or $\leq$ for missing fraction (Figure 3). In particular, A1 changes from 5.374856 μ S, above 4.74108975 μ S, to 4.214302 μ S, below it. Availability remains true while b_{A1} changes true to false. The admitted witness lists therefore give $(S(C_A), S(C_B), S(R)) : (1, 1, 1) \rightarrow (0, 1, 1)$.

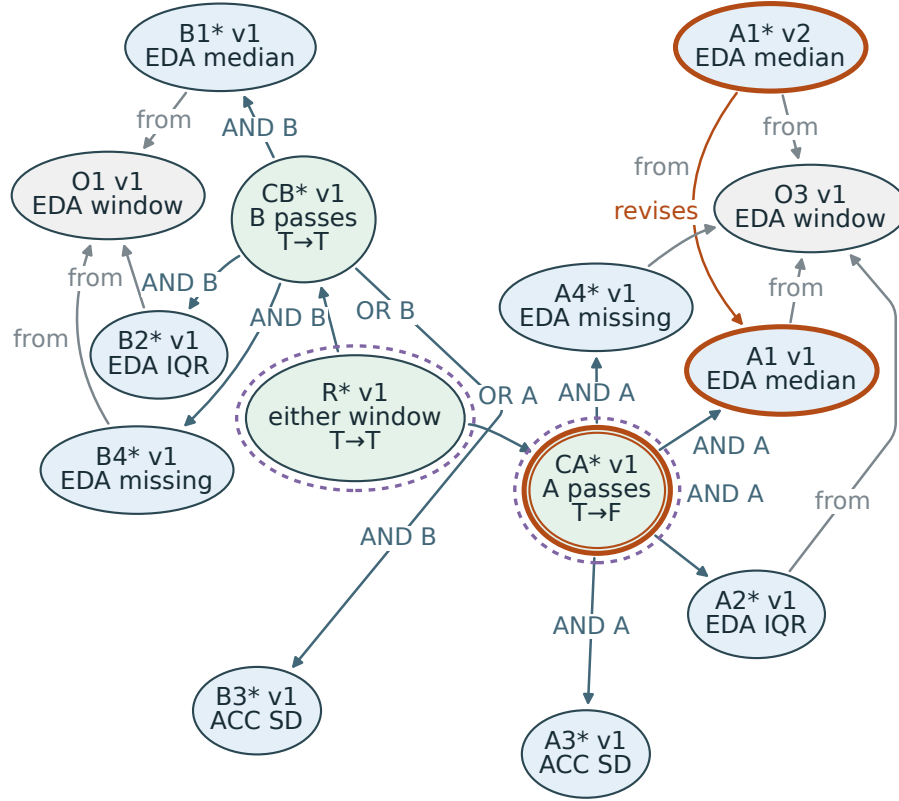
The theorem’s dependency projection has 13 nodes and 10 edges; provenance, membership and support mirrors do not enter its reachability. Equation 6 gives $U_\delta = \{A1_{v1}, A1_{v2}\}$ and $R_\delta = U_\delta \cup \{C_A, R\}$, but $B_\delta = \{C_A\}$. These are full computed sets, all visible here. The independent event/task oracle and recorded B2 schedule separately give $A_\delta = D_\delta = \{C_A\}$; Equation 7 is $0/1 = 0$. Reachability proposes two dependent claims for re-evaluation, yet the alternative witness preserves the answer. Direct and full maintenance accordingly agree for this update.

5.2 Grounded Content and Useful Coverage

Before/arrival/maintenance emitted-claim counts are Synthetic 211/211/138; WESAD 281/281/207; PPG-DaLiA 271/271/191. Answerable required-fact counts are 260/290/290 per source. Precision excludes empty outputs; recall includes their zero coverage. These pooled denominators describe the sample, while reported percentages average episodes within subjects.

The checked displays are fully grounded under the bounded numerical protocol, but they do not recover all useful information. Raw M1 required recall is 95.0%, 87.0% and 88.8% for synthetic, WESAD and PPG-DaLiA respectively; displayed recall falls to 78.2%, 75.8% and 75.5% (Table 3). The precision gain therefore includes rejection of correct useful propositions, not merely successful repair. Figure 5 isolates maintenance: re-evaluation restores grounding of retained content by withdrawing stale assertions, while required coverage can remain unresolved.

The contract crosswalk sharpens this interpretation. Of 799 B0 claims invalid under the frozen contract, 499 are grounded under the new proposition



○ ClaimVersion ○ FeatureVersion ○ Observation

* in current core K; unstarred nodes are review context

Orange: input versions U δ ; dotted: reachable claims

Double outline: changed support B δ ; labels: before $\rightarrow$ after

Full claim sets: reachable 2; changed 1; $\xi\delta = 0/1 = 0$

Fig. 4. Classic ontology-instance view of the admitted program `ppg_dalia-S1-e0-d2-alternative`, PPG-DaLiA S1 episode 0: controlled primary-route correction at 2331.0 s, inspected at 2331.01 s against 2301.01 s. All 14 ellipses are stored records. AND A/B label the four jointly required inputs of CA/CB; OR A/B label R's alternative parent witnesses. These arrows mean `DEPENDS_ON`; *from* means `DERIVED_FROM`, and *revises* means new-to-old `SUPERSEDES`. Original citations remain intact. Motion windows, administrative relations, feature dependency duplicates and `SUPPORTS` mirrors are hidden. Fill identifies ontology class; outlines and stars encode computed revision sets and core membership. No visual annotation is a database node or an additional dependency.

Table 3. Independent semantic scores in percent (subject means and 95% intervals). Each row contains 240 answer instances from 20 episodes. Matched counts unique requirements within each answer; denominators count answerable requirements. Empty displays have zero recall on answerable questions and undefined precision. The original strict-contract interpretation is preserved separately from these scores.

Source	State	P [95% CI]	R [95% CI]	Matched	Empty
Synthetic	B0 display	87.8 [84.5, 90.5]	95.5 [90.0, 99.5]	402/420	1
Synthetic	M1 raw	87.3 [83.4, 90.4]	95.0 [88.5, 100.0]	400/420	0
Synthetic	M1 display	100.0 [100.0, 100.0]	78.2 [66.2, 90.0]	333/420	20
WESAD	B0 display	83.6 [77.2, 89.4]	86.5 [77.8, 95.0]	366/420	1
WESAD	M1 raw	83.4 [77.1, 89.2]	87.0 [78.2, 95.5]	368/420	0
WESAD	M1 display	100.0 [100.0, 100.0]	75.8 [64.0, 87.5]	323/420	20
PPG-DaLiA	B0 display	84.5 [78.7, 89.6]	87.8 [79.2, 94.8]	371/420	1
PPG-DaLiA	M1 raw	84.7 [79.2, 89.6]	88.8 [80.7, 95.5]	375/420	0
PPG-DaLiA	M1 display	100.0 [100.0, 100.0]	75.5 [65.0, 87.5]	322/420	20

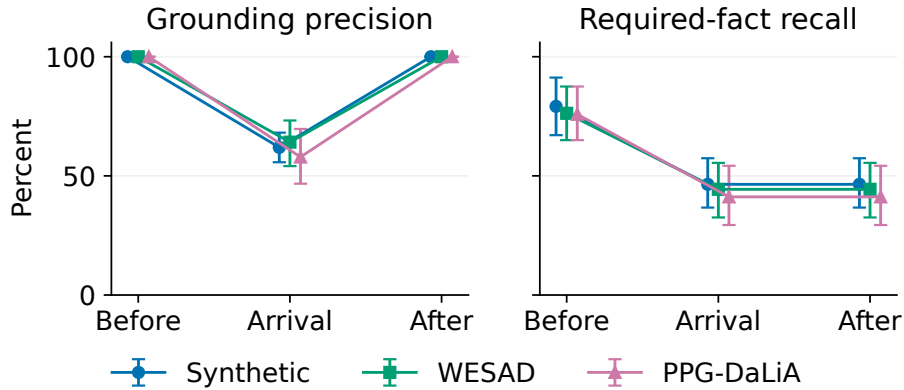


Fig. 5. Independent grounding precision and required recall for the same prior M1 answers at three discrete checkpoints, with subject-cluster 95% intervals. Each source has 160 transitions from 20 episodes and ten subjects across the four variants. No smooth recovery curve is inferred. Emitted-claim and required-fact denominators are given in the text; fresh-answer scores remain in Table 3.

protocol; 333 are correctly time-labelled background observations. Other disagreements include dependency/citation admission restrictions despite correct bounded numerical content. No claim passing the old contract becomes numerically ungrounded in this crosswalk. This does not turn an inaccurate target attribution into a correct background fact: matching uses the asserted interval and independently resolved version.

Source-window and known-version competency queries each agree with the independent log oracle on 1201 queried feature witnesses; comparison operand queries agree on 18 eligible displayed comparisons. These results test recoverability of admissible witnesses, rather than equality to a unique gold reasoning graph. The separate prose audit and original auxiliary-verifier disagreements remain available; bounded proposition scores do not certify unrestricted interpretation.

5.3 Structural Variation Tests the Maintenance Condition

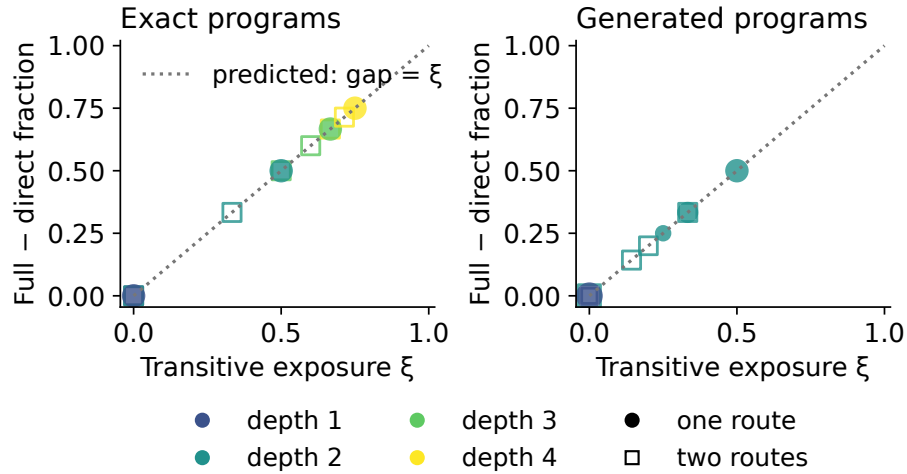


Fig. 6. Population-level structural test. Full-minus-direct correction fraction versus independently computed transitive exposure, separately for exact compiled and admitted GPU-generated programs. Points aggregate equal exposure, support regime and realized depth; size reflects batch count. The diagonal is the conditional prediction. This reports primary admission; the text gives the separate normalization sensitivity result.

The extension completes 240 common candidates with 325 GPU calls, including 85 repairs, in 17.2 minutes. Its 9,600 method/update replays include exact and generated structures. Neo4j and the relational control agree in all 1,920 matched origin/update groups. Direct maintenance misses 720 of 1260 required changes in exact programs and 183 of 1210 in admitted generated programs; full

Table 4. Common-candidate structural replay. Counts aggregate four update conditions per case, separately for exact and generated programs. A required change is independently determined before scheduling. Correction intervals average episodes within dataset-scoped subjects. B1 matches B2 and B3 matches M1 in every cell. Missing generated roots are included in the depth/coverage analysis, not invented as initially displayed claims.

Program	Method	Required	Missed	Collateral	Corrected % [CI]
Exact	B0	1260	1260	0	0.0 [0.0, 0.0]
Exact	B2	1260	720	0	42.9 [42.9, 42.9]
Exact	M1	1260	0	0	100.0 [100.0, 100.0]
Generated	B0	1210	1210	0	0.0 [0.0, 0.0]
Generated	B2	1210	183	0	84.9 [84.0, 85.7]
Generated	M1	1210	0	0	100.0 [100.0, 100.0]

maintenance misses none and causes no collateral withdrawal. The predicted missed set agrees in all 860 of 860 eligible groups (Figure 6).

Realized structure differs from the requested structure. Admitted generated answers have depth one in 74 cases and depth two in 135; 31 cases have no admitted answer root. None reaches depth three or four. The model commonly groups tests into route claims followed by an answer, without expressing the requested longer chain. Exact programs consequently test the deeper counterexamples, while generated programs test the dependencies the model actually expressed. Under primary-route loss, all 120 exact and 96 generated two-route answers with admitted roots retain support; single-route roots do not. Figure 3 makes preservation inspectable: the numerical update changes A1 from 5.3749 to 4.2143 μ S, below its 4.7411 μ S threshold. CA is withdrawn, but CB and the answer remain displayed. Direct and full maintenance agree for this route-loss update; necessary-change conditions in the population test expose excluded dependents. Necessary intermediate dependencies and surviving alternatives, rather than path length alone, determine maintenance behavior. In 236 paired cases, programs with the same evidence and baseline record count have different exposure, showing why total storage size is insufficient.

An admission-table omission was discovered after the core completed: primitive test names were defined in the input but unavailable as claim proposition names. It generated 236 unknown-proposition rejection reasons. A separate corrected replication normalizes those names to their supplied comparators without changing candidates, witnesses or links. It admits 229 roots and repeats 4,800 CPU database cells: direct maintenance misses 208 of 1294 required changes, while full maintenance again misses none. The original extension scores above remain frozen. Independent truth-table scoring distinguishes initially true statements from sound, complete witness families, so the admission correction is not treated as new model reasoning.

5.4 Operational Cost and Adequate Replacement

The original systems workload produces 0 adequate replacements among 236 completed candidates. All omit the required preceding-window difference. Their successful-replacement times therefore remain unresolved through observation. At the highest offered rate, both databases accumulate backlog and delayed flags. The extension isolates maintenance of a common answer; it adds no post-correction generation and hence establishes selective withdrawal and preservation, not automatic completion of a new adequate answer. Structural replay timings include diagnostic traversal and whole-program evaluation; they are instrumented costs, not production latency estimates.

6 Discussion and Conclusion

The representation distinguishes a changed value, an invalid prerequisite and a surviving witness. Its ontology makes these commitments auditable, and its review view exposes the relevant evidence and versions. Locality bounds candidate re-evaluation; transitive exposure identifies required changes excluded by direct scheduling. The two measured regimes show why declared paths and requested depth alone cannot establish a maintenance advantage.

Grounding and useful coverage remain separate. Withdrawal can remove stale content without restoring a required fact, and a strict contract can reject correct background information. Independent propositions and witnesses measure these distinctions. Admission normalization changes which supplied definitions the checker recognizes without changing model outputs or parent links.

The conclusions concern bounded propositions, fixed witness universes and acyclic dependencies. Recording agreement is not physiological truth. Natural faults, clock expiry, open-ended evidence discovery and unrestricted prose need further study. Ten held-out subjects per source, reuse of five for the extension, one generator and one seed limit generalization; disjoint windows are not statistically independent samples. Neo4j and the relational control agree on matched updates, without establishing a universal storage-performance ranking. The dashboard implements event-driven inspection and isolated review actions; human decision accuracy, interaction latency and real-time benefit were not evaluated. The original failure to produce adequate replacements limits claims of recovery.

Artifacts. The repository preserves frozen studies, raw candidates, independent scores, database exports, shared review views, vector figures and reproduction commands. All essential definitions and sensitivity results are in this paper; exhaustive logs remain repository data. Provider recordings are not redistributed.

Use of generative AI. An AI coding assistant assisted implementation, analysis and drafting. Qwen3-8B generated candidates; Qwen2.5-3B-Instruct supplied the auxiliary audit. Human authors remain responsible for reviewing evidence and interpretations before submission.

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